



Machine Learning Summary & Revision

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Terms



- Supervised & unsupervised learning
- Regression, classification
- Hypothesis, class
- Generalization
- Noise
- Inductive bias
- Underfitting, overfitting
- Bias, variance
- More complex & less complex hypothesis
- Regularization
- Evaluating model performance

More terms



Model parameters and hyperparameters

Parameters

- Required by the model when making predictions.
- Estimated or learned from data
- May not be set manually by the practitioner
- Saved as part of the learned model

Hyperparameters

- Used in processes to help estimate model parameters
- Specified by the practitioner
- They can often be set using heuristics
- Tuned for a given predictive modeling problem

Lazy and Eager learners

Lazy Learners

- Delays the process of modeling the training data until it is needed to classify the test examples
- No model building, but expensive to classify test examples
- Eg: kNN

Eager Learners

- Learns a model that maps the input attributes to the class label as soon as the training data becomes available
- Builds models, fast classification of test examples
- Eg: Decision tree, rule based classification, Naïve Bayes, SVM

More terms

- Generative models/functions

- Use prior class probabilities and apply Baye's theorem to find posterior probabilities
- Then determine class membership for each new input
- Learns the joint probability distributions of the data wrt to the class
- Assume some functional form for $P(Y)$, $P(X | Y)$
- Estimate parameters of $P(X | Y)$, $P(Y)$ directly from training data
- Use Bayes rule to calculate $P(Y | X)$
- E.g., Naive Bayes which directly estimates parameters for $P(Y)$ and $P(X | Y)$

- Discriminative models/functions

- Approaches that model posterior probabilities
- First solve the inference problem of determining the posterior class probabilities
- Then subsequently assign each new input to one of the classes
- Assume some functional form for $P(Y | X)$
- Estimate parameters of $P(Y | X)$ directly from training data
- Eg: Logistic regression (learning $P(Y|X)$ directly)



Linear Models for Regression

- Derive the gradient descent expression for multivariate linear regression, with regularization ✓
- What is the effect of regularization on model fitting? ✓
- How does large values of regularization parameter result in simpler hypothesis? ✓ λ θ_1, θ_2 very low
- Why cannot we use the MSE function as the cost function for logistic regression? ✓
- Derive an expression for Gradient descent rule for logistic regression with regularization

Linear Models for Classification



- Consider the hypothesis function $h(x) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2$ with learnt $w = \langle w_0, w_1, w_2, w_3, w_4 \rangle = \langle -36, 0, 0, 4, 9 \rangle$. What is the equation and shape of the decision boundary $g(x_1, x_2)$ for logistic regression given by
- $P(Y=1 | X) = 1/(1+e^{(h(x))})$?

$$P(Y=1 | X) = \underline{\underline{0.5}}$$

$$\frac{1}{1+e^{(h(x))}} = 0.5$$

$$e^{h(x)} = 1$$

$$\underline{\underline{h(x) = 0}}$$

$$h(x) = -36 + 0 + 0 + 4x_1^2 + 9x_2^2$$

$$4x_1^2 + 9x_2^2 - 36 = 0$$

$$\textcircled{\otimes} \quad \underline{\underline{4x_1^2 + 9x_2^2 = 36}}$$

Ellipse



Question 1

The sales of a company (in million dollars) for each year are shown in the table below.

- a) Find the least square regression line $y = \underline{a}x + \underline{b}$. ✓
b) Use the least squares regression line as a model to estimate the sales of the company in 2012. ✓✓✓

x (year)	2005	2006	2007	2008	2009
y (sales)	12	19	29	37	45

Solution 1



$$y = ax + b$$

$$a = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

$$b = \frac{1}{n} \left(\sum_{i=1}^n y_i - a \sum_{i=1}^n x_i \right)$$



Question 2

Consider the loss function of linear regression given by: $J(\theta_0, \theta_1)$.
Given $(\theta_0, \theta_1) = (0, 0.5)$, Estimate $\partial J / \partial \theta_1$ using the data points below:

x	2	4	7.0	8.0	10.0
y	1	2	2.5	3.5	5.5

Solution 2

Consider the loss function of linear regression given by: $J(\theta_0, \theta_1)$. Given $(\theta_0, \theta_1) = (0, 0.5)$, Estimate $\partial J / \partial \theta_1$ using the data points below:

x	2	4	7.0	8.0	10.0
y	1	2	2.5	3.5	5.5

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i) \cdot x_i$$

$$h_{\theta}(x_i) = \theta_0 + \theta_1 \cdot x_i$$

$$h_{\theta}(x_i) = 0 + 0.5 \cdot x_i = 0.5 \cdot x_i$$

So, for each x_i , the value of $h_{\theta}(x_i)$ is:

- For $x_1 = 2$: $h_{\theta}(2) = 0.5 \times 2 = 1$
- For $x_2 = 4$: $h_{\theta}(4) = 0.5 \times 4 = 2$
- For $x_3 = 7$: $h_{\theta}(7) = 0.5 \times 7 = 3.5$
- For $x_4 = 8$: $h_{\theta}(8) = 0.5 \times 8 = 4$
- For $x_5 = 10$: $h_{\theta}(10) = 0.5 \times 10 = 5$

x	Y	h(x)	h(x)-y	(h(x) - y)*x
2	1	1	0	0
4	2	2	0	0
7	2.5	3.5	1	7
8	3.5	4	0.5	4
10	5.5	5	-0.5	-5

Sum = 6

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{5} \times 6 = 1.2$$



Question 3

Vijay is a certified Data Scientist and he has applied for two companies -Google and Microsoft. He feels that he has a 60% chance of receiving an offer from Google and 50% chance of receiving an offer from Microsoft. If he receives an offer from Microsoft, he has belief that there are 80% chances of receiving an offer Google.

- If Vijay receives an offer from Microsoft, what is the probability that he will not receive an offer from Google?
- What are his chances of getting an offer from Microsoft, considering he has an offer from Google?

Question 4



Q.1 Consider the following dataset with 4 records.

Input X	Output Y
1	$\exp(2)$
2	$\exp(4)$
3	$\exp(6.3)$
4	$\exp(9.2)$

$$\ln y = \ln(e^{\alpha x})$$

$$\ln y = \alpha x$$

$$y_{\ln} = \alpha x$$

Assume output $y = e^{(\alpha * x)}$. Using linear regression,

(a) Find the best value of α .

(b) Find the optimal total sum of square error.

y	$y_{\ln} = \ln(y)$
e^2	$\ln(e^2) = 2$
e^4	$\ln(e^4) = 4$
$e^{6.3}$	6.3
$e^{9.2}$	9.2

Solution 4



To make this problem more suitable for linear regression, we take the natural logarithm of both sides of the equation $y = e^{\alpha x}$:

$$\ln(y) = \ln(e^{\alpha x}) = \alpha x$$

Q.1 Consider the following dataset with 4 records.

Input X	Output Y	$\ln y$	$y = e^{\alpha x}$	$\ln \hat{y} = \alpha x$	diff
1	exp(2)	2	e^{α}	α	$\alpha - 2$
2	exp(4)	4	$e^{2\alpha}$	2α	$2\alpha - 4$
3	exp(6.3)	6.3	$e^{3\alpha}$	3α	$3\alpha - 6.3$
4	exp(9.2)	9.2	$e^{4\alpha}$	4α	$4\alpha - 9.2$

Assume output $y = e^{(\alpha * x)}$. Using linear regression,

- Find the best value of α .
- Find the optimal total sum of square error.

Question 5

The 2-input AND gate is implemented using logistic regression classifier with gradient descent optimization algorithm. The model parameters at time t are given by $\theta_0=0$, $\theta_1=0$, and $\theta_2=0$. Given binary input (x_1, x_2) ,

- What will be value of the loss function at t ?
- What will be the values of θ_0 , θ_1 and θ_2 at $(t+1)$ with learning rate $\alpha=1$ and L2 regularization constant $\lambda=1$?

Data

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

Solution 5



(a)

✓✓ Cross entropy loss:

✓✓
$$= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

x1	x2	Target y	Predicted Output-yhat	y.ln(yhat)+(1-y)ln(1-yhat)
0	0	0	0.5	$0 \cdot \ln 0.5 + (1-0) \cdot \ln(1-0.5)$
0	1	0	0.5	$0 \cdot \ln 0.5 + (1-0) \cdot \ln(1-0.5)$
1	0	0	0.5	$0 \cdot \ln 0.5 + (1-0) \cdot \ln(1-0.5)$
1	1	1	0.5	$1 \cdot \ln(0.5)$

total loss=	0.693147181
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Solution 5



(b)

Cost function

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)}))] + \frac{\lambda}{2m} \sum_{j=1}^n \theta_j^2$$

Apply gradient descent update rule

y-hat	y	yhat-y	x0	(yhat-y)x0	w0-new
0.5	0	0.5	1	0.5	-0.25
0.5	0	0.5	1	0.5	
0.5	0	0.5	1	0.5	
0.5	1	-0.5	1	-0.5	

y-hat	y	yhat-y	x1	(yhat-y)x1	regularized w1-new
0.5	0	0.5	0	0	0
0.5	0	0.5	0	0	
0.5	0	0.5	1	0.5	
0.5	1	-0.5	1	-0.5	

y-hat	y	yhat-y	x2	(yhat-y)x1	regularized w2-new
0.5	0	0.5	0	0	0
0.5	0	0.5	1	0.5	
0.5	0	0.5	0	0	
0.5	1	-0.5	1	-0.5	

Question 6

- Suppose we train a model to predict whether an email is Spam or Not Spam. After training the model, we apply it to a test set of 500 new email messages (also labelled) and the model produces the contingency table below.
- [A] Compute the precision of this model with respect to the Spam class.
- [B] Compute the recall of this model with respect to the Not Spam class.

Predicted Class	True Class	
	Spam	Not Spam
	Spam	Not Spam
Spam	70	30
Not Spam	70	330

Solution 6

- Suppose we train a model to predict whether an email is Spam or Not Spam. After training the model, we apply it to a test set of 500 new email messages (also labelled) and the model produces the contingency table below.
- [A] Compute the precision of this model with respect to the Spam class.
- [B] Compute the recall of this model with respect to the Spam class.

		True Class	
		Spam	Not Spam
Predicted Class	Spam	70	30
	Not Spam	70	330

Answer:

[A] Precision with respect to SPAM =
 $\frac{\text{\# correctly predicted as SPAM}}{\text{\# predicted as SPAM}}$
 $= \frac{70}{70+30} = 70\%$

[B] Recall with respect to SPAM =
 $\frac{\text{\# correctly predicted as SPAM}}{\text{\# true SPAM}}$
 $= \frac{70}{70 + 70} = 50\%$

Question 7

Feature probability distribution of two classes of data points are given by

$$P(x, y | \text{class}_0) = \frac{1}{2\pi\sigma_x\sigma_y} \exp \left[-\frac{(x-1)^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2} \right]$$

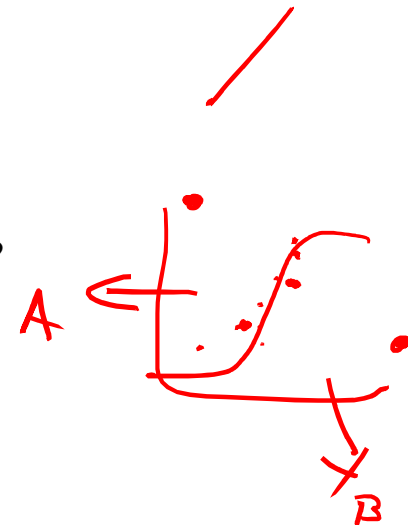
and

$$P(x, y | \text{class}_1) = \frac{1}{2\pi\sigma_x\sigma_y} \exp \left[-\frac{x^2}{2\sigma_x^2} - \frac{(y-2)^2}{2\sigma_y^2} \right]$$

Both classes are equally likely.

- Derive the equation of decision boundary.
- What is the equation and shape of the boundary if $\sigma_x = \sigma_y = \sigma$?

Derive the equation of decision boundary.



$$P(x, y | \text{class}_0) = P(x, y | \text{class}_1)$$



Solution 7

Feature probability distribution of two classes of data points are given by

$$P(x, y | \text{class}_0) = \frac{1}{2\pi\sigma_x\sigma_y} \exp \left[-\frac{(x-1)^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2} \right]$$

and

$$P(x, y | \text{class}_1) = \frac{1}{2\pi\sigma_x\sigma_y} \exp \left[-\frac{x^2}{2\sigma_x^2} - \frac{(y-2)^2}{2\sigma_y^2} \right]$$

Both classes are equally likely.

- a) Derive the equation of decision boundary.
- b) What is the equation and shape of the boundary if $\sigma_x = \sigma_y = \sigma$?

Derive the equation of decision boundary.

[ANS]

$$\frac{(x-1)^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2} = \frac{x^2}{\sigma_y^2} + \frac{(y-2)^2}{\sigma_x^2}$$

What is the equation and shape of the boundary if $\sigma_x = \sigma_y = \sigma$?

[ANS]

$$\frac{(x-1)^2}{\sigma^2} + \frac{y^2}{\sigma^2} = \frac{x^2}{\sigma^2} + \frac{(y-2)^2}{\sigma^2}$$

$$\text{Or, } -2x+1 = -4y+4$$

$$\text{Or } 2x - 4y = 3$$

$$\text{Or, } 2y - x = 1.5$$



Question 8

Consider these data points $\langle x, y \rangle$; $\langle 0, 2 \rangle, \langle 1, 3 \rangle, \langle 2, 5 \rangle, \langle 3, 4 \rangle, \langle 4, 6 \rangle$. Consider a linear hypothesis $h(x) = w^T x$. Answer the following questions:

- a) What is the equation of sum of squared error $J(w)$?
- b) Draw $J(w)$
- c) Calculate the global minima w_{min} and $J(w_{min})$
- d) Estimate the value of y when $x = 10$.

Solution 8

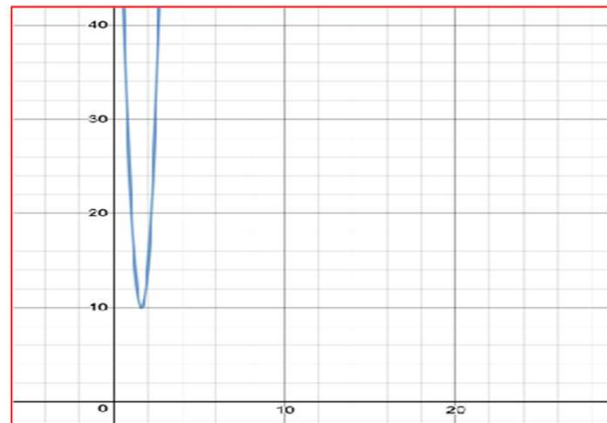
a) What is the equation of sum of squared error $J(w)$?

$$J(w) = \sum_i (wx_i - y_i)^2$$

$$= -2^2 + (w-3)^2 + (2w-5)^2 + (3w-4)^2 + (4w-6)^2$$

$$= 30w^2 - 98w + 90$$

b) Draw $J(w)$



c) Calculate the global minima w_{min} and $J(w_{min})$

$$J'(w) = 60w - 98$$

$$w_{min} \text{ when } J'(w) = 0$$

$$w_{min} = 1.633$$

$$J(w_{min}) = 30(1.633)^2 - 98(1.633) + 90$$

$$= 9.967$$

d) Estimate the value of y when $x = 10$.

$$\hat{y} = w_{min} \times x$$

$$= 1.633 \times 10 = 16.33$$